

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 5-6
9 February – 22 February 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



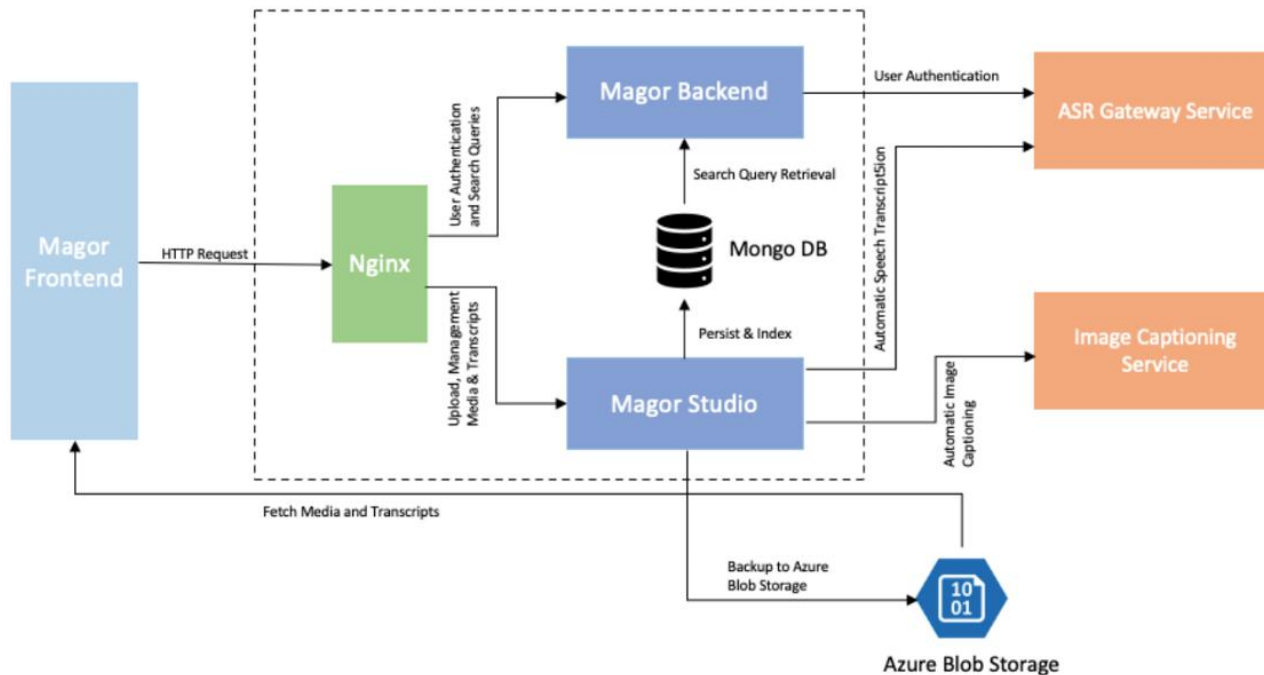
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Fixed TextGrid conversion (xmin correction)
2. Added multi-select options for transcript downloads
3. Implemented transcript download in Transcript Studio
4. Fixed video playback caption bugs



Task 1 (Fixed TextGrid conversion)

- Set $x_{\min}$ to 0

```
e-1770095240637.TextGrid
File Edit View

File type = "ooTextFile"
Object class = "TextGrid"

xmin = 0.09
xmax = 229.98
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "channell_1"
    xmin = 0.09
    xmax = 229.98
    intervals: size = 11
    intervals [1]:
      xmin = 0.09
      xmax = 34.63
      text = "oil and business sharp declines for India
      triggering a 30 minutes circuit breaker. The Jakarta Composi
      investability. DNS provider wants regulators to address conc
      wealthy owners. It's warning of a possible downgrade if prog
      under sustained pressure for months, tumbling to an all time
    intervals [2]:
      xmin = 34.63
      xmax = 39.15
      text = ""
    intervals [3]:
      xmin = 39.15
      xmax = 70.46000000000001

Ln 1, Col 1 5,248 characters Plain text
```



```
e-1770095240637 (2).TextGrid
File Edit View

File type = "ooTextFile"
Object class = "TextGrid"

xmin = 0
xmax = 229.98
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "channel1_1"
    xmin = 0
    xmax = 229.98
    intervals: size = 12
    intervals [1]:
      xmin = 0
      xmax = 0.09
      text = "--EMPTY--"
    intervals [2]:
      xmin = 0.09
      xmax = 34.63
      text = "oil and business sharp declines for Indone
triggering a 30 minutes circuit breaker. The Dakarta Composite
investability. DNS provider wants regulators to address concern
wealthy owners. It's warning of a possible downgrade if progre
under sustained pressure for months, tumbling to an all time l
intervals [3]:
  xmin = 34.63
```



Task 2 (Multi-select + Download All transcripts)

speech video test Edit Transcript ▾

Transcript ▾ Download ▾ Save [Go to Magor Transcript Studio](#)

00:00:00	have been rol	XML	
00:00:02	Resources.	STM	
00:00:03	for sustainab	TXT	
00:00:05	Ah, severely challenged.	SRT	
00:00:08	and many countries.	DOCX	
00:00:09	Really struggle.	TextGrid	
00:00:11	To make ends meet.		
00:00:13	Navigate.		



speech video test Edit Transcript ▾

Transcript ▾ Download ▾ Save [Go to Magor Transcript Studio](#)

00:00:00	have been rol	☒ Select All	
00:00:02	Resources.	☒ XML	
00:00:03	for sustainab	☒ STM	
00:00:05	Ah, severely c	☒ TXT	
00:00:08	and many cot	☒ SRT	
00:00:09	Really struggle.	☒ DOCX	
00:00:11	To make ends meet.	☒ TextGrid	
00:00:13	Navigate.	Download Selected (6)	

Task 2 (Multi-select + Download All transcripts)

The screenshot displays the Magor web application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a dropdown menu set to 'This recording'. Below the header, the page title 'speech video test' is visible, along with an 'Edit Transcript' button. The main content area features a video player on the left and a transcript list on the right. The video player shows a man speaking at a podium, with a progress bar at the bottom indicating '0:00 / 0:13'. The transcript list on the right has a 'Download' button highlighted with a mouse cursor. The transcript entries are as follows:

Transcript	Download	Save
00:00:00	have been rolled back.	
00:00:02	Resources.	
00:00:03	for sustainable development.	
00:00:05	Ah, severely challenged.	
00:00:08	and many countries.	
00:00:09	Really struggle.	
00:00:11	To make ends meet.	
00:00:13	Navigate.	

Task 3 (Download Transcripts in Transcript Studio)

Magor

test

Title

speech video test

Description

No Description

Go to Recording



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Transcripts

Sort By Created Date (Earliest) ▾

Transcript ID: 6980f554438c1d0023c35ef4

Created At: 16 days ago

Last Updated At: 16 days ago

Auto Generated

Edit

Delete

Transcript ID: 6980f596438c1d0023c35f08

Created At: 16 days ago

Last Updated At: 16 days ago

Auto Generated

Edit

Delete



Magor

test

Title

speech video test

Description

No Description

Go to Recording



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⬇

Transcripts

Sort By Created Date (Earliest) ▾

Transcript ID: 6980f554438c1d0023c35ef4

Created At: 16 days ago

Last Updated At: 16 days ago

Auto Generated

Edit

Download ▾

Delete

Transcript ID: 6980f596438c1d0023c35f08

Created At: 16 days ago

Last Updated At: 16 days ago

Auto Generated

Edit

Download ▾

Delete



Task 3 (Download Transcripts in Transcript Studio)

localhost/transcriptStudio/6980f4fb438c1d0023c35ee0

Magor

test

Title

speech video test

Description

No Description

Go to Recording



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Transcripts

Sort By Created Date (Earliest) ▾

Transcript ID: 6980f554438c1d0023c35ef4	Created At: 16 days ago	Last Updated At: 16 days ago
Auto Generated		
Edit	Download ▾	Delete
Transcript ID: 6980f596438c1d0023c35f08	Created At: 16 days ago	Last Updated At: 16 days ago
Auto Generated		
Edit	Download ▾	Delete



Task 4 (Fixed video playback caption bugs)

- Bug 1: Multiple sentences in current caption slot

The screenshot shows a video player interface for a video titled "Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4". The video shows two men in a studio setting. The player interface includes a search bar, a "This recording" dropdown, and a "test" button. Below the video, there are tags, top words, and a transcript. The transcript shows a list of sentences, with the first sentence highlighted in a red box, indicating a bug where multiple sentences are displayed in a single caption slot.

Magor Search This recording test

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 Edit

Tags: CNA

Top Words: party parliament opposition leader workers pritam singh lo en time position people office mister

Transcript Show options

channel1_S3 Applies equally to the PAPM Ps.

channel1_S1 yup

channel1_S3 And especially the political office holders. Right, so, so there will be, you know, that so-called blow back or you come back to haunt them.

channel1_S3 Right. You know, if the PAP doesn't Um Uh, doesn't conform. you know, to those high standards, right? Or, or they be accused of double standards?

channel1_S2 and,

0 matches and I think we will not be able to, be I think it was something that was necessary, right? Because the political office holders. Right, so, so there will be, you know, that so-called blow back or you come back to haunt them.

28:07 / 30:42

Task 4 (Fixed video playback caption bugs)

- Bug 2: Incorrect lines displayed

Magor Search This recording test

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 Edit

Tags: CNA

Top Words: party parliament opposition leader workers pritam singh lo en time position people office mister

Transcript Show options

channel1_S3 Applies equally to the PAPM Ps.

channel1_S1 yup

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channel1_S2 and,

and I think we will not be able to, be

I think it was something that was necessary, right? Because

Right, so, so there will be, you know, that so-called

Right, so, so there will be, you know, that so-called

Right. You know, if the PAP doesn't

28:09 / 30:42

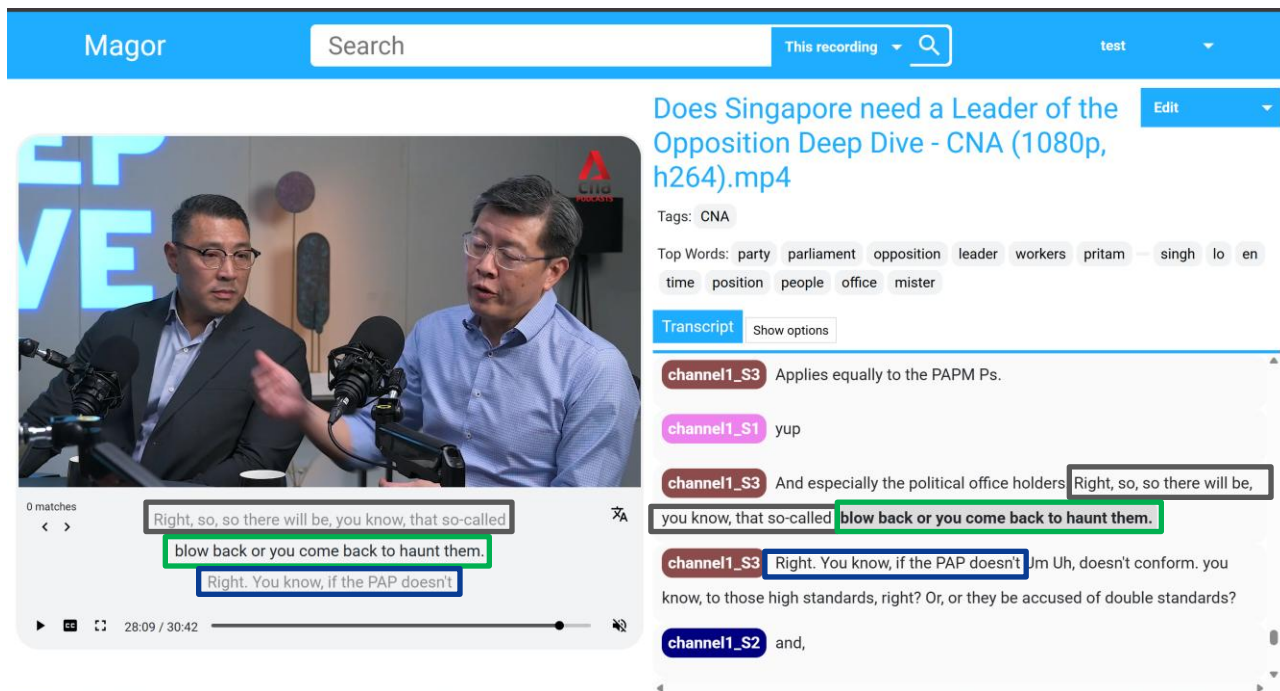
Repeated lines

Incorrect captions



Task 4 (Fixed video playback caption bugs)

- Solution: Timeline checking and proper clearing of captions



Magor Search This recording test

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 Edit

Tags: CNA

Top Words: party parliament opposition leader workers pritam singh lo en time position people office mister

Transcript Show options

channel1_S3 Applies equally to the PAPM Ps.

channel1_S1 yup

channel1_S3 And especially the political office holders Right, so, so there will be, you know, that so-called blow back or you come back to haunt them.

channel1_S3 Right. You know, if the PAP doesn't Jm Uh, doesn't conform. you know, to those high standards, right? Or, or they be accused of double standards?

channel1_S2 and,

0 matches

Right, so, so there will be, you know, that so-called blow back or you come back to haunt them.

Right. You know, if the PAP doesn't

28:09 / 30:42

Plans for Next Week

- Further investigate video playback issues
 - Multiple lines of transcript highlighted and incorrect caption order
 - Navigation dropdown unable to open when video is playing
- Compress multiple file downloads into a ZIP file
- Add docx library to package json

